

Day 16 – AWS Auto Scaling & Load Balancing

Introduction

AWS Auto Scaling automatically adjusts EC2 instances based on traffic demand.

Benefits

Improves availability, handles traffic spikes, and optimizes cost.

Load Balancing

Elastic Load Balancer distributes traffic across multiple EC2 instances.

Types of Load Balancers

ALB, NLB, and Gateway Load Balancer.

Scaling Policies

Scale Out adds instances; Scale In removes instances.

Real-World Project

Scalable web application with Auto Scaling and Load Balancer.

Security Features

IAM Roles, Security Groups, HTTPS traffic.

Monitoring

CloudWatch tracks CPU, latency, request count, and EC2 health.

Advantages

Fault tolerant, scalable, highly available.

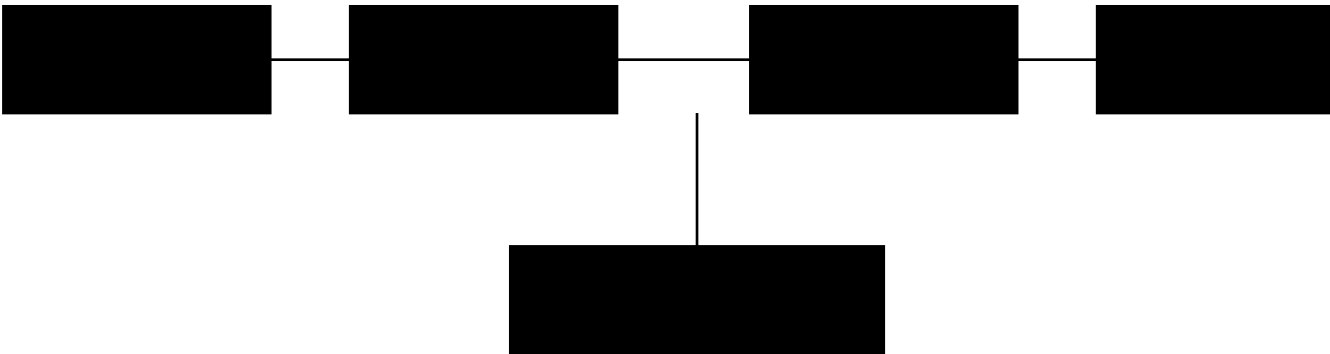
Use Cases

E-commerce, banking, gaming, SaaS applications.

Hands-On Tasks

Create EC2 template, configure ALB, setup Auto Scaling Group.

Auto Scaling Workflow



AWS Components Table

Service	Purpose
Auto Scaling	Automatically adjusts EC2 instances
Load Balancer	Distributes traffic
CloudWatch	Monitoring and alarms
EC2	Application servers

Detailed Workflow

Users ↓ Load Balancer ↓ Auto Scaling Group ↓ Multiple EC2 Instances

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